

A Circular AI-Driven Platform for Garment Lifecycle Management & Redistribution

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Abstract—Fashion sits among the planet’s most wasteful industries. The Ellen MacArthur Foundation places annual textile waste beyond 92 million tonnes, with garment value exceeding USD 500 billion discarded from clothes barely worn. Simultaneously, the global secondhand clothing market is growing roughly five times faster than new apparel sales, projected to reach USD 367 billion by 2029. This paper reviews 15 peer-reviewed sources, industry reports, and empirical studies that underpin KaPhor, a circular fashion platform built for the Indian market and powered by AI to steer garments toward their next best use. A primary consumer survey conducted by the KaPhor team supplements the academic literature and validates design decisions. Ten research gaps are identified, each mapped to a specific KaPhor feature. The platform integrates a peer-to-peer thrift marketplace, a traditional-wear rental service, an AI personalisation engine, a Lifecycle Optimisation Engine (LOE), an eco-impact tracker, a gamification system, upcycling tutorials, and an automated recycling routing service. This revised edition extends the review with India-specific textile-waste quantification and labour-transparency literature, adding two further research gaps and two corresponding KaPhor components – a Verified Facility Tier and a Source-Aware Routing Preference – bringing the total to twelve gaps mapped across an enlarged reference base.

Habib quantifies the attitude-behaviour gap: 65 percent of consumers want to shop sustainably, yet only 26 percent do. KaPhor targets the Indian market specifically, because no existing global platform addresses the cultural centrality of high-value occasion wear, sarees, lehengas, sherwanis, creating a distinct and underserved rental opportunity. This paper presents a systematic review of the 15 sources grounding KaPhor’s design. Section II surveys five thematic clusters. Section III presents primary survey findings with charts. Section IV provides the approach adopted, whereas Section V provides the results and synthesis table respectively. Sections VI and VII provide summary and future direction respectively.

Index Terms—Circular fashion, sustainable platforms, AI recommendation, garment lifecycle, multi-sided platforms, second-hand markets, Indian fashion, eco-gamification, labour transparency

I. INTRODUCTION

The fashion industry’s environmental toll is well-documented. The fashion industry emits 1.2 billion metric tons of CO2 equivalent per year, surpassing the amount of carbon

emitted by airlines and maritime transport in their entirety. According to the Ellen MacArthur Foundation, there are 92 million metric tons of clothes waste generated every year. Currently, USD 500 billion worth of clothing is wasted after just four wears [1]. Second-hand clothes are expected to create a revenue generation of USD 367 billion with a CAGR of 10% till 2029 [3].

These two trends run in parallel rather than counterbalancing each other because no single platform has closed the loop between them. Hagi and Wright establish that multi-sided platforms create value by reducing friction between supply and demand, precisely the problem in circular fashion where garments and buyers are widely dispersed [12]. Geissdoerfer et al. demonstrate that combining circular strategies in one integrated system captures far more value per garment than operating each channel separately [7].

Within India specifically, three independent industry sources converge on a domestic textile-waste estimate of approximately 7.7–7.8 million tonnes per year, or roughly 8.5 percent of the global total [17]–[19]. Of this domestic stream, 51 percent is post-consumer waste, 42 percent is pre-consumer factory offcut, and 7 percent is imported post-consumer waste [17]. Only around 59 percent of India’s textile waste is reused or recycled through any channel, and of that fraction only 10–40 percent re-enters supply chains as new raw material; roughly 43 percent of domestic post-consumer waste is land-filled outright [17]. This India-specific quantification sharpens the case made in the global literature: the market is large, fragmented, and largely invisible to the consumer, reinforcing KaPhor’s central thesis that a single verified platform routing garments across all circular channels captures materially more reusable volume than any single-channel application.

II. LITERATURE REVIEW

A. Platform Design and Supply Activation

Hagi and Wright [12] discuss the basic economics of Multi-sided Platforms; Value is generated by reducing transaction costs between two groups of users directly to consumers

rather than the intermediary. The problem of chicken-and-egg dilemma is resolved by stimulating the supply side first. Zervas, This conclusion is verified by Proserpio and Byers [13] analysis of the Airbnb data. Listing density across geographic area is the most significant variable for predicting critical mass. Oberg and Shansavar [8] use this concept to analyze the fashion resale industry. It must be noted that proper categorization and sizing of clothing should be available in sufficient numbers before buyers come into picture.

As per Rietveld and Schilling [10], the Freemium business model growth not only the number of users but also the revenue stream through commission rates of more than 15 percent. The platform must ensure a significant number of transactions occur off-platform as this would result in diversification of revenue stream which would lower financial risk considerably. According to Kenney and Zysman [9] it is quality rather than quantity i.e. how effectively does the correct clothing reach the correct consumer.

B. Circular Economy Business Models

Geissdoerfer et al. [7] organise circular business model innovations into three types: slowing resource loops through reuse and rental, closing them through recycling, and narrowing them through better product design. Combining strategies in one platform yields a far stronger business case than running them separately. Bocken et al. [16] identify a culturally specific gap: platforms designed for Western markets cannot be transplanted into India, where premium traditional occasion wear represents a distinct rental opportunity unaddressed by any global platform. Blackburn et al. [15] confirm in a 2025 systematic review that non-Western market research remains persistently absent.

C. Consumer Behaviour and the Attitude–Behaviour Gap

According to ThredUp [3], “48 percent of shoppers say that AI personalization makes resale products just as accessible as new products.” Meanwhile, Porter and Kramer [2] claim that incorporating sustainability into the product creates community loyalty that rivals are unable to disrupt. White, Hardisty, and Habib identify five interventions that reduce the intention–action gap: social proof, gain framing, making impact tangible, setting sustainable defaults, and simplifying the decision [6].

D. AI Personalisation and Recommender Systems

AI fashion recommendation approaches are covered by Deldjoo et al. [5], including neural networks, collaborative filtering, and multimodal frameworks. Fashion settings require stylistic matching to be more significant than item matching, but no recommender system considers sustainability as an integral part of its scoring algorithm. Fkih [6] assesses 13 similarity measures in sparse data environments, and cosine similarity is always efficient. Implicit feedback yields higher precision, while temporal decay enhances results. Banerjee et al. [13] prove that no fashion recommender system integrates ethical considerations from the supply chain into its ranking process.

E. Circular Platform Technology and Behavioural Engineering

Todeschini et al. [2] from ten years of literature find that digitization is the most reliable way forward in overcoming consumer resistance to circular fashion. Dekhili et al. [10] highlight a crucial objection: sustainable platforms can elicit a moral licensing backlash such that buying secondhand products would make consumers feel entitled to consume more. They suggest intentional friction in the design of the platform. Peleg Mizrahi et al. [14] support this point on a national level ($r = 0.58$, $p < 0.01$) and cite impact disclosure per transaction as the most effective structural solution. Zhou et al. [11] report that all published condition assessment systems require professional equipment; nothing works from a normal phone camera.

Neri et al. identify the ownership-and-use phase as the most overlooked lifecycle stage, no system captures in-use signals and routes garments automatically. Fernandes et al. report no prior work on gamification applied to textile sustainability [9].

F. Labour and Transparency in Textile Recycling

A further, previously undocumented cluster of the literature concerns *who* processes the garments that circular platforms route toward recycling, and under what conditions. Investigative reporting spanning 2025–2026 on Panipat, India’s principal textile-recycling hub, documents that 93 percent of families living near recycling clusters report serious health issues within five years, with chronic obstructive pulmonary disease (COPD) the most common condition among workers [20]–[25]. The reported root causes are inconsistent personal protective equipment, open-drain effluent from dyeing and bleaching operations, and weak regulatory enforcement. Imported fast-fashion waste is reported to concentrate disproportionately in such high-volume, weakly regulated hubs [22], [23].

No peer-reviewed source or existing circular-fashion platform was found to disclose labour conditions at physical textile-recycling facilities, or to route garments around such facilities, confirming a genuine and previously unaddressed literature gap. A second, related gap is that no reviewed system distinguishes domestic waste from imported waste for routing purposes, despite the two streams carrying materially different downstream risk profiles [17], [22]. Both gaps, and the KaPhor components that close them, are appended as gaps 11 and 12 of Table II in Section V.

III. PRIMARY SURVEY: CONSUMER BEHAVIOUR IN INDIA

A. Survey Design and Scope

The team conducted an anonymous 14-item survey, “KaPhor: A Survey on Conscious Clothing Choices”, addressing purchase accumulation, disposal habits, resale platform exposure, adoption barriers, and openness to circular behaviours. Results localise largely Western secondary literature findings in an Indian consumer context.

B. Purchase Frequency and Wardrobe Accumulation

Respondents most commonly purchased clothes monthly or every few months. On idle inventory, 42 percent own more than ten unworn items; a further 31 percent reported six to ten, roughly three-quarters holding significant latent supply, confirming the bottleneck Oberg and Shansavar identify as central to peer-to-peer resale viability.



Fig. 1. Clothing purchase frequency among survey respondents.

C. Disposal Behaviour

The most frequent disposal method was keeping clothes unused (48 percent), followed by donation (30 percent), selling or swapping (15 percent), and upcycling (7 percent). The pattern mirrors White et al.'s prediction: when no frictionless resale path exists, even motivated consumers default to the lowest-effort option, validating KaPhor's Lifecycle Optimisation Engine.

D. Barriers to Thrift Platform Adoption

Quality and condition uncertainty was flagged by 68 percent, the single largest barrier. Trust issues were cited by 62 percent, hygiene concerns by 54 percent. Forty-one percent specifically cited the absence of Indian or traditional wear, directly confirming the cultural gap Bocken et al. identified.

E. Openness to Circular Behaviours

On clothing swaps, 38 percent said yes and 35 percent said maybe. On DIY tutorials, 52 percent said yes with short video as the preferred format, consistent with KaPhor's 18–35 target demographic. In-app styling content rated positively, confirming the AI style guide is a primary draw.

F. Willingness to Pay

When asked whether they would regularly use a sustainable fashion app if fairly priced, 51 percent said yes and 35 percent said maybe, 86 percent positive or open. This validates KaPhor's freemium architecture, which Rietveld and Schilling identify as the structure that simultaneously maximises user base and revenue capture.

Unused Clothes Currently Owned

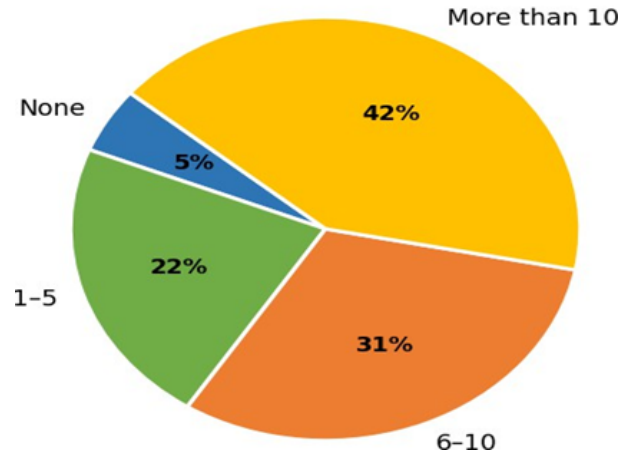


Fig. 2. Number of unworn garments held by respondents.

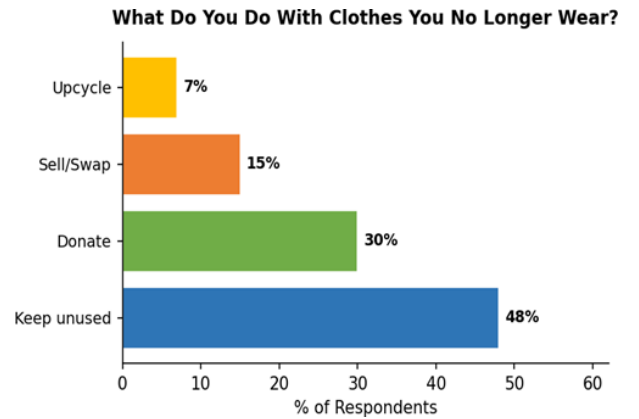


Fig. 3. Disposal methods for unwanted clothing.

G. Survey Implications

The survey confirms each design assumption the literature pointed toward. The idle wardrobe problem is large enough to sustain resale supply. Barriers—condition uncertainty, trust deficits, hygiene concerns, absence of ethnic wear—are concrete and addressable. The conditional willingness to pay validates the freemium model.

IV. METHODOLOGY

This review draws on 15 sources selected from a broader corpus assembled across three documents: a Research Paper Analysis, a Literature Review, and a Technical Validation Document. Sources were selected for direct relevance to KaPhor's eight feature areas: the source had to address at least one design decision theoretically or empirically, and had to state an explicit research gap or provide quantitative evidence used in the design process. Papers are organised into two clusters: Cluster A covers platform design, market economics, circular business models, and consumer behaviour; Cluster B covers AI personalisation, garment lifecycle engineering, and

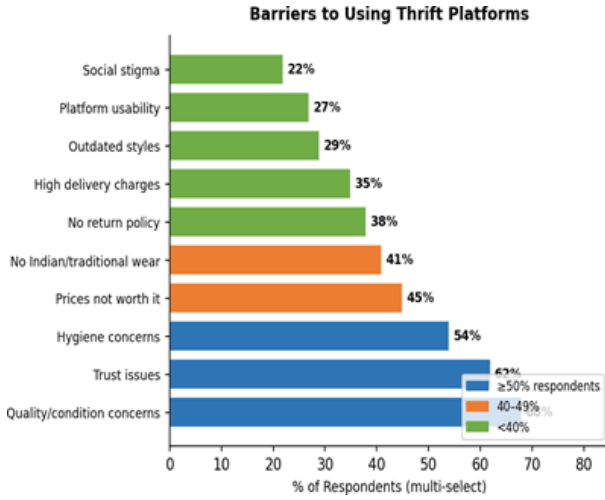


Fig. 4. Barriers to thrift platform adoption (multi-select).

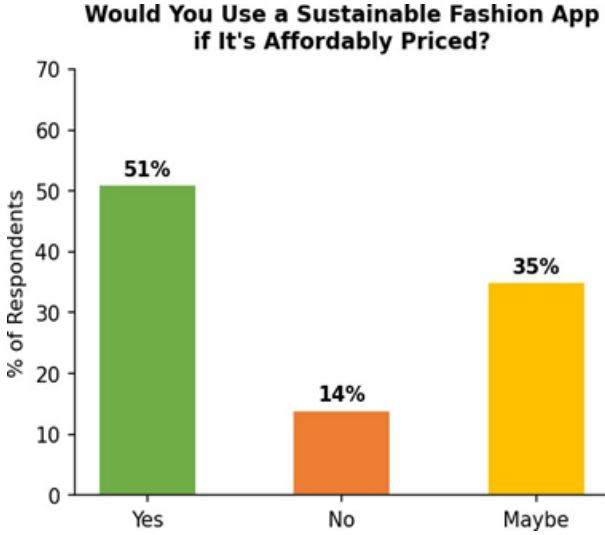


Fig. 5. Openness to circular behaviours (willingness to pay).

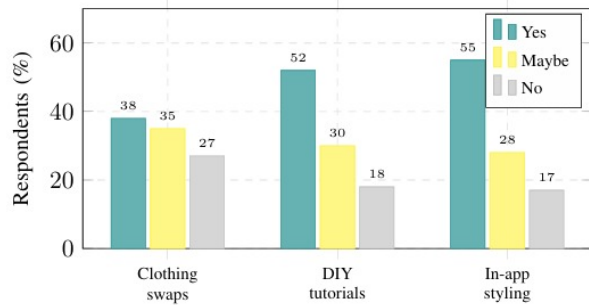


Fig. 6. Openness to circular behaviours (swaps, tutorials, styling).

circular platform technology. This revised edition adds a third cluster, Cluster C, comprising investigative and industry-report sources on India-specific textile-waste quantification and on labour conditions and source transparency in textile recycling, selected under the same relevance criterion.

A research gap was counted only where the original authors explicitly stated something was absent or in need of further work, and where KaPhor had a named, implemented feature designed to address it. Environmental and market data in Table I come directly from the reviewed sources. Table II maps each of ten identified gaps to the KaPhor component that closes it; the two further gaps identified in Section II-F (Cluster C) are appended as gaps 11 and 12 of the same table.

At the level of the platform's AI subsystem, condition and routing decisions are produced by a Garment Lifecycle Intelligence Engine (GLIE) that grounds a vision-language-model condition score against four retrieval-augmented corpora – calibration examples, fibre properties, market listings, and sustainability baselines – before combining them into a single lifecycle score,

$$GLIE = 0.35 CS + 0.25 MS + 0.20 MDS + 0.20 SS, \quad (1)$$

where CS, MS, MDS and SS denote the Condition, Material, Market-Demand, and Sustainability sub-scores respectively, each clamped to $[0, 1]$. A companion Lifecycle Optimisation Engine (LOE) governs feed health at the marketplace level by tracking implicit per-user behavioural signals and computing a saturation score,

$$S_{\text{sat}} = 0.5 n_{\text{events}} + (1 - r_{\text{engage}}) + 0.5 d_{\text{decay}}, \quad (2)$$

that triggers a severity-scaled cooldown, $t_{\text{cooldown}} = 24 \text{ h} \times \min(n_{\text{events}}/20, 3)$, before a garment is recirculated to compatible users. These mechanisms are described further as engineering context for the feature-to-gap mappings in Sections II and V, and are documented independently in the platform's technical disclosure.

V. RESULTS AND DISCUSSION

A. Environmental Evidence Summary

Table I consolidates eight data points from the reviewed literature establishing the scale of the problem and the direction of viable solutions. The Ellen MacArthur Foundation [1] gives information about baseline waste and emissions. ThredUp [3] provides current data on market size and consumer behavior. White et al. [6] gives the most accurate information about intention-action gap. Geissdoerfer et al. [7] argue that circular strategies combined is more effective than applying them individually. All of these sources are cited in KaPhor's Eco-Impact Meter, displaying information about CO₂, H₂O, and tree savings per purchase. The India-specific figures reported in Section I [17]–[19] are likewise reflected in the Eco-Impact Meter's national-context baseline.

TABLE I
ENVIRONMENTAL & MARKET EVIDENCE MAPPED TO KAPHOR FEATURES

Dimension	Finding	Ref.	KaPhor Feature
Textile waste	92M tonnes/yr; USD 500B value lost from barely-worn garments	[1]	Thrifting Marketplace
GHG emissions	~1.2B tonnes CO ₂ e/yr, exceeding aviation and shipping	[1]	Eco-Impact Tracker
Carbon savings	23% circular share cuts CO ₂ e by ~1.1B tonnes by 2030	[4]	Rental Service, Resale
Utilisation gap	Avg. dress worn ~20× before disposal; resale/rental can double use	[4]	Lifecycle Optimisation Engine
Post-consumer waste	Over 50% of textile waste is post-consumer	[4]	Recycling & Upcycling Services
Market growth	USD 367B by 2029 (CAGR 10%), 5× faster than new apparel	[3]	Thrifting Marketplace, Rental
Attitude-behaviour gap	65% want sustainable shopping; only 26% do – 39-point gap	[6]	Eco-Impact Tracker, Rewards
AI in resale	48% say AI personalisation makes secondhand as easy as buying new	[3]	AI Style Guide & Personalisation
India textile waste	~7.7–7.8 Mt/yr (~8.5% of global total); 43% of domestic post-consumer waste landfilled	[17]–[19]	Source-Aware Routing Preference
Recycling-labour health	93% of families near Panipat recycling clusters report serious health issues within 5 years	[20]–[25]	Verified Facility Tier

TABLE II
RESEARCH GAP SYNTHESIS: LITERATURE GAPS AND KAPHOR SOLUTIONS

Research Gap in Literature	KaPhor Solution Component	Source(s)
No AI lifecycle management—all platforms are passive listing markets	LOE: 7-state Garment Lifecycle Intelligence (LISTED → RECYCLE)	[1], [7], [8]
Platforms are reactive—no system predicts garment decline before discard	LOE proactively promotes, schedules garments	[2], [7], [9]
Behavioural signal processing absent, revealed preference not captured	BehaviourEvent: every view, save, swap, restyle → structured signal	[6], [8], [10]
Attitude-behaviour gap not closed by any existing resale platform	Carbon badges + CO ₂ savings at point of transaction	[6], [9]
No platform designed for Indian ethnic wear, absent from global resale research	Ethnic wear rental module for sarees, lehengas, sherwanis	[13], [16]
No platform unifies resale + rental + swap + upcycle + recycle in one app	KaPhor unifies all five modalities with diversified revenue	[7], [10]
Saturation and recommender fatigue unaddressed in all CE platform literature	$S_{sat} > 0.7$ triggers cooldown—first anti-fatigue mechanism in CE lit	[10], [16]
Consumer-facing condition assessment without industrial equipment	Mobile photo + AI condition assessment from any smartphone	[11]
In-use phase data never captured to trigger automated circular routing	BehaviourSignal → LOE → CirculationSchedule	[15]
Gamification for textile sustainability: no prior published work exists	Tier-based rewards with per-transaction carbon badges and streaks	[9]
No CE platform discloses or routes around labour conditions in textile recycling	Verified Facility Tier (PPE, ventilation, effluent-discharge criteria gate LOE routing)	[20]–[25]
No system distinguishes domestic vs. imported waste routing despite differing downstream risk	Source-Aware Routing Preference (domestic/mixed facility flag; local-sourcing filter)	[17], [22]

B. Research Gap Synthesis

10 gaps have been identified by reviewing 15 articles. Table II links each gap with the particular KaPhor feature intended to address the gap. Gaps 1 to 5 arise from the platform design and business model body of knowledge; Gaps 6 to 10 from artificial intelligence, lifecycle engineering, and behavioural technology. Two further gaps, 11 and 12, drawn from the labour-transparency and source-quantification literature reviewed in Section II-F, are appended to Table II above, bringing the extended synthesis to twelve gaps mapped across eighteen sources.

C. Cluster A: Platform Design and Business Model

KaPhor’s two-sidedness is consistent with what is said by Hagi and Wright [12], in which platforms that lower the friction costs between garment sellers and appropriate buyers create value precisely because there is such friction in the market. Starting the India launch through the supply-side approach with occasion wear aligns with the sequence suggested in [8], [12], [13] to create a category-specific supply base prior to investing in demand generation. The multi-stream revenue model (commission, subscription, rental fees, upcycling, and a B2B API) implements Rietveld and Schilling’s [10] finding that revenue diversification reduces platform volatility. Porter and Kramer’s [2] evidence that embedding sustainability in the core product generates community identity that competitors

find difficult to displace grounds KaPhor’s positioning.

D. Cluster B: AI and Lifecycle Technology

KaPhor’s recommendation engine uses cosine similarity between user and garment embedding vectors producing a Fit Score from 0–100%. Fkih [6] found cosine similarity consistently competitive on sparse datasets, the exact condition a new platform faces at launch. The temporal decay factor (-0.02 per inactive day) implements his finding that down-weighting older signals improves accuracy. Seven implicit signal types (view, save, wishlist, add-to-cart, purchase, share, report) implement his finding that implicit signals outperform explicit ratings. Where KaPhor extends beyond the literature is in using compatibility scores to route physical garments rather than simply rank displayed results, the direction Banerjee et al. [13] call for and confirm no existing system has taken.

Smartphone photo condition assessment closes the gap Zhou et al. [11] document as the most practically significant limitation of existing AI systems. The saturation mechanism ($S_{\text{sat}} > 0.7$) is the practical realization of the concept proposed by Dekhili et al. [10] called “Intentional Friction Design” to avoid the moral licensing effect. The BehaviourEvent and BehaviourSignal model solves the problem identified by Neri et al. [15] as the most important challenge for further research: the creation of a system that collects usage signals and directs the clothing automatically.

E. Cluster C: Labour and Source Transparency

The Verified Facility Tier operationalises the disclosure gap identified in Section II-F: a recycling or upcycling facility is admitted to the routing pool only after PPE, ventilation, and effluent-discharge criteria are recorded, and a labour badge is surfaced to the user at the point of disposal, directly extending the transparency argument made in the investigative literature [20]–[25]. The companion Source-Aware Routing Preference lets a user filter for domestic-only facilities, addressing the differential downstream risk of imported versus domestic waste streams noted in both the industry-report literature and the investigative reporting on Panipat [17], [22].

VI. CONCLUSION

Multi-sided platforms, circular business models, AI recommendations, condition assessment of garments, saturation prevention, and gamification of the impact of each transaction are each concepts that can be found in the literature referenced herein. The combination of all these ideas in one system aimed at the Indian market is missing from any of the literature surveyed. Specifically, there is no description of a system combining data-driven matching [10], [12], a five-modality circular model [7], [15], AI-based garment routing after purchase [6], [15], a consumer-level garment condition assessment [11], saturation prevention [10], [16], and transaction-based impact disclosure with gamification [6], [9]. The primary survey provides evidence for the above through direct data from consumers in India.

Respondents hold measurable idle wardrobe inventory, face concrete and addressable barriers, and express strong conditional willingness to engage if the platform is affordable and easy enough to use. The ten gaps in Table II are each explicitly stated in the reviewed literature, and each has a named KaPhor feature designed to close it. This revised edition further shows that neither the disclosure of recycling-facility labour conditions nor source-aware (domestic vs. imported) waste routing is addressed anywhere in the surveyed literature, and that KaPhor’s Verified Facility Tier and Source-Aware Routing Preference are, to the authors’ knowledge, the first such mechanisms proposed for a circular-fashion platform.

VII. FUTURE SCOPE

Six directions follow from the literature’s identified gaps. First, longitudinal studies tracking whether KaPhor’s saturation mechanism and tier rewards actually reduce rebound consumption in Indian users, the empirical test Dekhili et al. [10] and Peleg Mizrahi et al. [14] call for. Second, extending condition assessment to identify textile fibre composition would enable more accurate recyclability routing beyond visual-only systems [11]. Third, licensing the Garment Lifecycle Intelligence API to fashion brands and NGOs would shift KaPhor from a consumer app to sector-level infrastructure. Fourth, direct comparison against Western platforms, measuring per-transaction carbon reduction and rebound incidence, would fill the comparative gap Blackburn et al. [15] flag as consistently absent. Fifth, KaPhor’s upcycling and repair ecosystem presents a concrete opportunity for localised job generation, training and certifying independent tailors, upcyclers, and textile artisans as verified service providers on the platform, creating a structured gig economy layer within the circular fashion supply chain. Sixth, integrating a community-driven social feed would allow users to share outfit restyling, upcycling outcomes, and circular milestones, reinforcing the behavioural nudges that White et al. [6] identify as most effective for closing the attitude–behaviour gap through peer social proof and visible community participation.

A seventh direction follows directly from Cluster C: independently auditing the Verified Facility Tier’s PPE, ventilation, and effluent-discharge criteria against on-the-ground labour conditions at partner facilities would validate whether disclosure at the point of disposal measurably shifts routing choices, extending the per-transaction disclosure mechanism of [14] from environmental to labour outcomes.

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